

5/4 Grp Mtg

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Compactified Jacobians of Singular curves

Goal For a sm proj curve C ,

$\text{Jac}(C) := \text{Pic}^0(C)$ abelian var.

which parametrizes degree zero line bundles on C .

Q what happens to this object when the curve becomes singular?

Q what breaks, and how can we repair it?

→ explain answers by analyzing two basic degenerations — nodal curves / cuspidal curves.

(describing their Picard var & compactifications explicitly).

§1 Picard group (just set $\mathcal{O}$)

geom str.

up to isom

X : curve

$\text{Pic}(X) := \{\text{line bundles on } X\} / \cong$ w/ $\otimes$

$$\begin{aligned} & \dim H^0(C, L) \\ & - \dim H^1(C, L) \\ & + g - 1 \end{aligned}$$

For curves $\ni$ natural inv: $\deg: \text{Pic}(C) \rightarrow \mathbb{Z}$, $\mathcal{O}_C(D) \mapsto \deg D$ sum of wts

decomposes

$$\text{Pic}(C) = \bigsqcup_{d \in \mathbb{Z}} \text{Pic}^d(C)$$

parametrizes

the bundle of degree d .

most imp part for us will be the deg zero part.

$\text{Pic}^0(C)$: Jacobian

two types of information

discrete inv: $\deg$ $\text{Pic}^0(C)$
continuous family: $\text{Pic}^0(C)$

when $X = \text{sm pvg}$ ~~etc~~. $\text{Pic}^0(X)$ has a very rich structure. ^[2]
 → see Birkenhake-Lange "Complex Abelian Varieties" Ch II

§2 Jacobian

Prop $C = \text{sm pvg}$ $\text{Pic}^0(C)$ is abelian var.

of $\dim g(C)$ mod over principally pol.

$$E \text{ ellip} \Rightarrow \text{Pic}^0(E) \cong E$$

$$\odot T_{\text{Pic}^0(C)} \cong H^1(C, \mathcal{O}_C)$$

by deformation theory

$$\text{by theta div} = \text{AJ}_{g-1}^{(g)}(C)$$

cpt & sm

A genus g curve has g independent directions to vary line bundles

it can be constructed from divisors on C

$$\text{AJ}_1: C \rightarrow \mathbb{A}^g / \Lambda \quad \left(\begin{array}{c} H^0(C, \mathcal{O}_C^*) \\ \int_{p_0}^p w_1, \dots, \int_{p_0}^p w_g \end{array} \right) \Big/ H^1(X, \mathbb{Z})$$

Abel map $C \mapsto \text{Pic}^0(C)$ $p_0 \in \text{fix}$
 $- \text{Jacobi}$ $p \mapsto \mathcal{O}(p - p_0)$

$$L(\mathcal{O}_{p_0}) \cong \mathcal{O}(D)$$

$$\left\{ \begin{array}{l} h^0(M) \geq 1 \\ \text{AJ}_1 = 0 \Leftrightarrow D: \text{principle} \end{array} \right\} \text{generically } h^0(M) = 1$$

$$\rightsquigarrow \text{Sym}^d C \mapsto \text{Pic}^0(C)$$

$$(p_1, \dots, p_d) \mapsto \mathcal{O}(p_1 + \dots + p_d - dp_0)$$

$$d \geq g \Leftrightarrow \text{surj} \quad \Downarrow$$

$$\text{AJ Div}^0 / \sim \cong \text{Alb}(C)$$

Picard variety behaves like a fine moduli space with a univ family

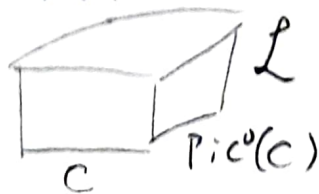
(we skip discussion about Picard functor, representability)

$$\text{Pic}^0_T = \text{Pic}^0(C \times_k T)$$

$$\exists L \in \text{Pic}(C \times \text{Pic}^0(C))$$

$$\text{s.t. (i) } L|_{C \times \{L\}} \cong L \text{ for every } L \in \text{Pic}^0(C)$$

$$(ii) L|_{\{L\} \times \text{Pic}^0(C)} \text{ is trivial normalization}$$



see Kleiman
 The Picard scheme in FGA explained

Now what happens when the curve requires singularities [3]
 We can ^{still} define $\text{Pic}^0(C)$ line bundle still make sense on sing curve.

However it's no longer an abelian var in general.
 typically becomes non-compact

C : integral (irreducible & reduced) $\implies$

$\nearrow \text{Pic}^0(C)$: gen. Jac $\xrightarrow{\text{deg: Pic}^0(C) \rightarrow \mathbb{Z}^{\# \text{comp}}}$
 $\nu: \tilde{C} \rightarrow C$ normalization multidegree

To avoid unnecessary complications arising from multiple components.

Today Altman-Kleiman (80) Compactifying the Picard scheme
 Oda-Seshadri (79) Compactifications of the generalized Jacobian var.

Also see Jesse Kass: ^{#17} Lecture notes on cpt'd Jac. & some surveys.

For non-reduced $\rightarrow$ Bosch-Lütkebohmert-Raynaud (90)

"New models"
^{#17, #18, #19, #20, #21, #22, #23, #24, #25, #26, #27, #28, #29, #30, #31, #32, #33, #34, #35, #36, #37, #38, #39, #40, #41, #42, #43, #44, #45, #46, #47, #48, #49, #50, #51, #52, #53, #54, #55, #56, #57, #58, #59, #60, #61, #62, #63, #64, #65, #66, #67, #68, #69, #70, #71, #72, #73, #74, #75, #76, #77, #78, #79, #80, #81, #82, #83, #84, #85, #86, #87, #88, #89, #90, #91, #92, #93, #94, #95, #96, #97, #98, #99, #100}
 Czapliński-Krug-Lehn-Pollenske (24)
 "cpt Jac of ext ADE curve"

We focus on

$\implies$ Geometry is governed by a single component?

$\text{Pic}^0(C)$ fit into an exact sequence

$$0 \rightarrow G \rightarrow \text{Pic}^0(C) \rightarrow \text{Pic}^0(\tilde{C}) \rightarrow 0$$

G : alg group determined by the singularities

The simplest examples already show two different behaviors.

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nodal curve : $G \cong G_m$ multiplicative parameter } extra factor.
cuspidal curve : $G \cong G_a$ additive parameter

$$\text{Pic}^0(C) \cong \text{Jac}(\tilde{C}) + \text{extra directions} \\ \Rightarrow \text{non cpt?}$$

Moreover, if we study limits of line bundles in a family of curves, the limit may fail to be locally free.

We enlarge the Picard var. by allowing rank 1 torsion free sheaves. $\Rightarrow$ compactified Jac

$\overline{\text{Pic}}^0(C)$. curve

e.g. X_0 : plane cubic $\vee$ in $\mathbb{P}^2$ w/ node at $p_0 = [0, 0, 1]$

X_∞ : general cubic $\vee$ passing thru p_0

$\{X_t\}$: pencil. $\{X_0 + tX_\infty\}_{t \in \mathbb{P}^1}$

general 1-dim lin system

general

$t \neq 0 \Rightarrow X_t$ sm

$t \neq 0 \Rightarrow$ we can define line bundle

$\mathcal{L}_t = \mathcal{O}_{X_t}(-p_0)$ ideal sheaf of p_0

natural ask what is limit?

$\lim_{t \rightarrow 0} \mathcal{L}_t = ? = \mathcal{I}_{p_0}$ on X_0

(this is not a line bundle?)

Basic example of rank 1 torsion free

{ line bundle
non zero ideal sheaf

$S_m \Rightarrow \mathcal{O}_{X,p} : \text{DVR (fingen) torsion free} \Rightarrow \text{free module}$
discrete valuation ring

[5]

node $\rightarrow$ paper

future plan

Today, we have focused on the behavior of Jac under degeneration.

This perspective is not only a technical necessity, but also provides a conceptual foundation for studying Lagrangian fibrations arising from families of curves.

In particular in the setting of BM system
or Deligne system

the geometry of the total space governed by
the cpt Jac of y can